Prompt - Generate MATLAB code that takes a video file as input using `VideoReader`.Ask the user whether they want to upsample or downsample the video.Ask the user to enter the scaling factor (e.g., 2× or 0.5×).Process and resize every video frame while maintaining the aspect ratio.Save the processed video as a new file using `VideoWriter`.

```
% video_resample.m
% Usage: run this script in MATLAB. It prompts for inputs and writes output
file.

% Ask for input video
[inputFile, inputPath] = uigetfile({'*.mp4;*.avi;*.mov;*.mkv','Video Files
(*.mp4,*.avi,*.mov,*.mkv)'; '*.*','All Files'}, 'Select input video');
if isequal(inputFile,0)
    disp('No file selected. Exiting.')
    return
end
inFull = fullfile(inputPath, inputFile);

% Create VideoReader and basic info
vr = VideoReader(inFull);
fprintf('Input: %s\nResolution: %dx%d  FrameRate: %.2f\n', inputFile,
vr.Width, vr.Height, vr.FrameRate);

% Select resampling type
type = lower(input('Resample type - spatial or temporal? (s/t): ','s'));
if isempty(type) || ~ismember(type(1),['s','t'])
    error('Invalid selection. Choose s or t.');
end

% Upsample or downsample
dir = lower(input('Direction - upsample or downsample? (u/d): ','s'));
if isempty(dir) || ~ismember(dir(1),['u','d'])
    error('Invalid selection. Choose u or d.');
end

% Factor input
factor = input('Enter resampling factor (e.g. 2 for double size or half
size): ');
if ~isnumeric(factor) || factor <= 0
    error('Factor must be numeric and > 0.');
end

% Prepare output filename
[~,name,ext] = fileparts(inputFile);
outName = sprintf('%s_%s_%s_f%.2g%s', name,
tern(type(1)=='s','spatial','temporal'), tern(dir(1)=='u','up','down'),
factor, ext);
```

```
outFull = fullfile(inputPath, outName);

% Configure VideoWriter
% For spatial changes keep same frame rate (unless temporal resampling
chosen)
if type(1) == 's'
    outFR = vr.FrameRate;
else
    % temporal will change frame rate if up/down sampling
    if dir(1) == 'u'
        outFR = vr.FrameRate * factor;
    else
        outFR = vr.FrameRate / factor;
    end
end
vw = VideoWriter(outFull, 'MPEG-4');
vw.FrameRate = outFR;
open(vw);

% Process frames
fprintf('Writing to %s (FrameRate=%.2f)\n', outName, outFR);
while hasFrame(vr)
    frame = readFrame(vr);
    if type(1) == 's'
        % Spatial resampling using imresize
        if dir(1) == 'u'
            scale = factor;
        else
            scale = 1/factor;
        end
        % Use bicubic for smooth scaling
        frameOut = imresize(frame, scale, 'bicubic');
        writeVideo(vw, frameOut);
```